

(1) (2) (3) (4) (5) (6) (7) (8) (9) (10) (11) (12) (13) (14) (15) (16) (17) (18) (19) (20) (21) (22) (23) (24) (25) (26) (27) (28) (29) (30) (31) (32) (33) (34) (35) (36) (37) (38) (39) (40) (41) (42) (43) (44) (45) (46) (47) (48) (49) (50) (51) (52) (53) (54) (55) (56) (57) (58) (59) (60) (61) (62) (63) (64) (65) (66) (67) (68) (69) (70) (71) (72) (73) (74) (75) (76) (77) (78) (79) (80) (81) (82) (83) (84) (85) (86) (87) (88) (89) (90) (91) (92) (93) (94) (95) (96) (97) (98) (99) (100)

MULTIVARIATE ANALYSIS.

06/01/2024

1D MULTIVARIATE NORMAL DISTRIBUTION

UNIVARIATE

X: Height



Random sample: $(x_1, x_2, \dots, x_n)$

MULTIVARIATE

X: Height

Y: Weight

Z: Age

Random sample: $(x_1, x_2, \dots, x_n)$

$$\underline{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

UNIVARIATE NORMAL DISTRIBUTION

$$X \sim N(\mu, \sigma^2)$$

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2 \right], \quad x \in \mathbb{R}, \mu \in \mathbb{R}, \sigma^2 > 0$$

$\underbrace{\frac{1}{\sigma \sqrt{2\pi}}}_{\text{constant}}$
 $\underbrace{\left(\frac{x-\mu}{\sigma} \right)^2}_{\text{Quadratic form}}$

Univariate: x, μ, σ^2

Multivariate: $\underline{x}, \underline{\mu}, \Sigma$

Now consider p variables $\underline{x} = (x_1, \dots, x_p)$

Let's construct the PDF of MVN distribution.

$$\begin{cases} \underline{\mu} \rightarrow \underline{b} \\ (\sigma^2)^{-1} \rightarrow A \end{cases} \begin{cases} \text{P.d. matrix} \\ \text{full rank} \end{cases}$$

$$(\underline{x}-\underline{\mu})(\sigma^2)^{-1}(\underline{x}-\underline{\mu}) \rightarrow (\underline{x}-\underline{b})' A (\underline{x}-\underline{b})$$

$$= \sum_i \sum_j a_{ij} (x_i - b_i) (x_j - b_j)$$

So, the PDF becomes -

$$f_{x_1, \dots, x_p}(x_1, \dots, x_p) = f_{\underline{x}}(\underline{x}) = k \cdot \exp \left[-\frac{1}{2} (\underline{x}-\underline{b})' A (\underline{x}-\underline{b}) \right]$$

From the properties of PDF -

$$k \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \exp \left[-\frac{1}{2} (\underline{x}-\underline{b})' A (\underline{x}-\underline{b}) \right] d\underline{x} = 1$$

Since, A is a positive definite matrix, so $\exists$ a non-singular matrix C s.t.

$$C'AC = I.$$

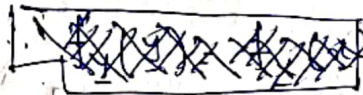
$$A = (CC')^{-1} \text{ or } A^{-1} = CC'$$

Transformation: $\underline{x} - \underline{b} = C\underline{y}$

$$\underline{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_p \end{pmatrix}$$

$$(\underline{x} - \underline{b})'A(\underline{x} - \underline{b}) = \underline{y}'C'AC\underline{y} = \underline{y}'\underline{y}$$

Jacobian of Transformation $|J| = |C|$



$$f_Y(\underline{y}) = K \exp\left[-\frac{1}{2} \underline{y}'\underline{y}\right] |C|$$

$$= K |CC'|^{1/2} \exp\left[-\frac{1}{2} \underline{y}'\underline{y}\right]$$

$$= K |A^{-1}|^{1/2} \exp\left[-\frac{1}{2} \underline{y}'\underline{y}\right]$$

$$\int_{\mathbb{R}^p} f_Y(\underline{y}) d\underline{y} = 1$$

$$\Rightarrow K \int_{\mathbb{R}^p} |A^{-1}|^{1/2} \exp\left[-\frac{1}{2} \sum_{i=1}^p y_i^2\right] d\underline{y} = 1$$

$$\Rightarrow K |A^{-1}|^{1/2} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \prod_{i=1}^p \exp\left(-\frac{1}{2} y_i^2\right) dy_i = 1$$

$$\Rightarrow K |A^{-1}|^{1/2} \prod_{i=1}^p \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2} y_i^2\right) dy_i = 1$$

$$\Rightarrow K |A^{-1}|^{1/2} \left(\sqrt{2\pi}\right)^p = 1$$

$$\Rightarrow K = \frac{1}{(2\pi)^{p/2} |A^{-1}|^{1/2}}$$

$$E(y_i) = \int \dots \int y_i \cdot \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} y_j^2\right) dy_1 \dots dy_p$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y_i \exp\left(-\frac{1}{2} y_i^2\right) dy_i \cdot \left(\prod_{j \neq i} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} y_j^2\right) dy_j \right)$$

$$= 0 \times 1 = 0$$

$$\therefore E(\underline{y}) = \underline{0} \quad \text{and} \quad E(\underline{x}) = \underline{b}$$

Therefore, $\underline{b} = \underline{\mu} = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_p \end{pmatrix}$

Now, $E(\underline{x} - \underline{b})(\underline{x} - \underline{b})' = c E(\underline{y} \underline{y}') c' = \underline{0}$

$$= c E \begin{pmatrix} y_1^2 & y_1 y_2 & \dots & y_1 y_p \\ y_2 y_1 & y_2^2 & \dots & y_2 y_p \\ \vdots & \vdots & \ddots & \vdots \\ y_p y_1 & y_p y_2 & \dots & y_p^2 \end{pmatrix} c'$$

Now, $E(y_i^2) = 1$ and $E(y_i y_j) = 0$ for $i \neq j$

$$= \left(\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} y_i^2 \exp\left(-\frac{1}{2} y_i^2\right) dy_i \right) \cdot \left(\prod_{j \neq i} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} y_j^2\right) dy_j \right)$$

$$\therefore E(\underline{x} - \underline{b})(\underline{x} - \underline{b})' = c c' = A^{-1}$$

$$\Rightarrow E(\underline{x} - \underline{\mu})(\underline{x} - \underline{\mu})' = A^{-1}$$

$$\Rightarrow \Sigma = A^{-1}$$

$$\Rightarrow \boxed{A = \Sigma^{-1}}$$

Therefore, we finally get the PDF as -

$$f_{\underline{x}}(\underline{x}) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left[-\frac{1}{2} (\underline{x} - \underline{\mu})' \Sigma^{-1} (\underline{x} - \underline{\mu}) \right]$$

$$\underline{x} \in \mathbb{R}^p$$

10/01/2024

Theorem > If the density of a p -dim RV $\underline{x}$ is

$$\frac{\sqrt{|A|}}{(2\pi)^{p/2}} \exp \left[-\frac{1}{2} (\underline{x} - \underline{b})' A (\underline{x} - \underline{b}) \right] \text{ then}$$

$$E(\underline{x}) = \underline{b} \quad \& \quad \text{cov}(\underline{x}) = A^{-1}$$

Conversely, given a $\underline{\mu}$ & Σ (p.d.) there is a

MVN distribution $\frac{1}{(2\pi)^{p/2}} \frac{|\Sigma|^{-1/2}}{\exp \left[-\frac{1}{2} (\underline{x} - \underline{\mu})' \Sigma^{-1} (\underline{x} - \underline{\mu}) \right]}$

$$\text{s.t. } E(\underline{x}) = \underline{\mu} \quad \& \quad \text{cov}(\underline{x}) = \Sigma$$

Remarks >

$$\underline{\theta} = (\theta_1, \dots, \theta_n)'$$

$f(\underline{\theta})$ is a real valued function of $\theta_1, \theta_2, \dots, \theta_n$

$$\frac{\partial f(\underline{\theta})}{\partial \underline{\theta}} = \begin{pmatrix} \partial f(\underline{\theta}) / \partial \theta_1 \\ \vdots \\ \partial f(\underline{\theta}) / \partial \theta_n \end{pmatrix}$$

e.g. If A is a $n \times n$ symmetric matrix & $f(\underline{\theta}) = \underline{\theta}' A \underline{\theta}$

$$\frac{\partial f(\underline{\theta})}{\partial \underline{\theta}} = (A + A') \underline{\theta} = 2A \underline{\theta}$$

→ Characteristic function >

Univariate > $\phi(\theta) = E[\exp(i\theta X)] = \int e^{i\theta x} f(x) dx$

Multivariate > $\phi(\underline{\theta}) = E[\exp(i\underline{\theta}' \underline{x})] = \int_{\mathbb{R}^p} e^{i\underline{\theta}' \underline{x}} f(\underline{x}) d\underline{x}$
($\underline{x} \in \mathbb{R}^p$)

Properties > → $\phi(0) = 1$

→ $|\phi(\underline{\theta})| \leq 1$

→ Uniqueness Theorem >

Two RVs have the same characteristic function iff they have the same distribution.

INVERSION THEOREM

Given a characteristic function, we can find the PDF by inversion theorem,

Differentiating the characteristic f.r. w.r.t. $\underline{\theta}$ and then put $\underline{\theta} = \underline{0}$.

Characteristic function

MVN Distribution

$$\underline{X} \sim N_p(\underline{\mu}, \Sigma)$$

$$\phi(t) = E[\exp(it' \underline{X})]$$

$$\text{Let, } \underline{X} - \underline{\mu} = \underline{u}$$

$$\text{Then, } \phi(t) = E[\exp[it'(\underline{u} + \underline{\mu})]] = \exp(it' \underline{\mu}) E[\exp(it' \underline{u})]$$

$$= \exp(it' \underline{\mu}) E[\exp(i(t' \underline{e}) \underline{y})]$$

$$= \exp(it' \underline{\mu}) E[\exp(i \underline{u}' \underline{y})]$$

$$= \exp(it' \underline{\mu}) \prod_{j=1}^p E[\exp(i u_j y_j)]$$

$$[\underline{X} \sim N(\underline{\mu}, \sigma^2), \phi_X(t) = \exp[i \underline{\mu}' t - \frac{1}{2} \sigma^2 t^2]$$

$$\underline{\mu} = 0, \sigma^2 = 1, \phi_Z(t) = \exp(-\frac{1}{2} t^2)$$

$$\text{Treat } u_j = t, y_j = Z. \text{ So, } E(e^{i u_j y_j}) = e^{-\frac{1}{2} u_j^2}$$

$$\text{So, } \phi(t) = \exp(it' \underline{\mu}) \prod_{j=1}^p \exp(-\frac{1}{2} u_j^2)$$

$$= \exp(it' \underline{\mu}) \cdot \exp(-\frac{1}{2} \underline{u}' \underline{u})$$

$$= \exp(it' \underline{\mu} - \frac{1}{2} \underline{u}' \underline{u}) \quad [\underline{u}' \underline{u} = t' \underline{C} t]$$

$$= \exp(it' \underline{\mu} - \frac{1}{2} t' \Sigma t) \quad = t' \Sigma t]$$

DISTRIBUTION OF LINEAR COMBINATION

$$\underline{X} \sim N_p(\underline{\mu}, \Sigma)$$

$$\underline{Z} = \underline{A} \underline{X}, \quad \underline{A} \text{ is not necessarily full rank.}$$

$$\phi_z(t) = E[\exp(it'z)]$$

$$= E[\exp(it'A'x)]$$

$$= E[\exp(i\theta'z)]$$

$$= \exp[i\theta'\mu - \frac{1}{2}\theta'\Sigma\theta]$$

$$= \exp[it'(A\mu) - \frac{1}{2}t'(A\Sigma A')t]$$

Using uniqueness theorem, z follows M.V.N. with mean vector $A\mu$ & cov. matrix $A\Sigma A'$.

$$z = Ax \sim N_p(A\mu, A\Sigma A')$$

11/01/2024

$$x = (x_1, x_2, \dots, x_p)' = (x_1', x_2')'$$

$$x \sim N_p(\mu, \Sigma)$$

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \rightarrow \begin{matrix} \gamma x_1 \\ (1-\gamma)x_1 \end{matrix}$$

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}_{(1-\gamma) \times (1-\gamma)}$$

To find the distribution of x_1 & x_2 .

[Remark: Say x_1 & x_2 are independent.
then $\Sigma_{12} = 0 = \Sigma_{21}$

$$\text{Then } \Sigma = \begin{pmatrix} \Sigma_{11} & 0 \\ 0 & \Sigma_{22} \end{pmatrix}$$

$$\therefore \Sigma^{-1} = \begin{pmatrix} \Sigma_{11}^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix} \text{ \& } |\Sigma| = |\Sigma_{11}| \cdot |\Sigma_{22}|$$

Now, the PDF of x is -

$$f_x(\mu, \Sigma) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp\left[-\frac{1}{2}(\underline{x} - \underline{\mu})' \Sigma^{-1} (\underline{x} - \underline{\mu})\right]$$

$$\theta = [(\underline{x}_1 - \underline{\mu}_1)', (\underline{x}_2 - \underline{\mu}_2)']' \begin{pmatrix} \Sigma_{11}^{-1} & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix} \begin{pmatrix} \underline{x}_1 - \underline{\mu}_1 \\ \underline{x}_2 - \underline{\mu}_2 \end{pmatrix}$$

$$= (\underline{x}_1 - \underline{\mu}_1)' \Sigma_{11}^{-1} (\underline{x}_1 - \underline{\mu}_1) + (\underline{x}_2 - \underline{\mu}_2)' \Sigma_{22}^{-1} (\underline{x}_2 - \underline{\mu}_2).$$

$$= Q_1 + Q_2.$$

$$f_{\underline{x}}(\underline{\mu}, \Sigma) = \frac{1}{(2\pi)^{q/2} |\Sigma_{11}|^{1/2}} \exp\left(-\frac{Q_1}{2}\right) \cdot \frac{1}{(2\pi)^{\frac{p-q}{2}} |\Sigma_{22}|^{1/2}} \exp\left(-\frac{Q_2}{2}\right).$$

$$= f_{\underline{x}_1}(\underline{\mu}_1, \Sigma_{11}) \cdot f_{\underline{x}_2}(\underline{\mu}_2, \Sigma_{22})$$

i.e. $\underline{x}_1 \sim N_q(\underline{\mu}_1, \Sigma_{11})$ & $\underline{x}_2 \sim N_{p-q}(\underline{\mu}_2, \Sigma_{22})$ independently

General case ($\Sigma_{12} \neq 0$)

$$\text{Choose, } \underline{y}_1 = \underline{x}_1 + B \underline{x}_2$$

$$\underline{y}_2 = \underline{x}_2$$

Select B s.t. $\underline{y}_1$ & $\underline{y}_2$ are uncorrelated.

$$\text{i.e. } E[(\underline{y}_1 - E(\underline{y}_1))(\underline{y}_2 - E(\underline{y}_2))'] = 0$$

$$E[(\underline{x}_1 - E(\underline{x}_1)) + B(\underline{x}_2 - E(\underline{x}_2))](\underline{x}_2 - E(\underline{x}_2))' = 0$$

$$\Rightarrow \Sigma_{12} + B \Sigma_{22} = 0$$

$$\Rightarrow B = -\Sigma_{12} \Sigma_{22}^{-1}$$

$$\therefore \underline{y} = \begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{I} & -\Sigma_{12} \Sigma_{22}^{-1} \\ 0 & \mathbf{I} \end{pmatrix} \begin{pmatrix} \underline{x}_1 \\ \underline{x}_2 \end{pmatrix} \quad [\text{Non-singular transformation}]$$

$$E \begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{I} & -\Sigma_{12} \Sigma_{22}^{-1} \\ 0 & \mathbf{I} \end{pmatrix} \begin{pmatrix} \underline{\mu}_1 \\ \underline{\mu}_2 \end{pmatrix} = \begin{pmatrix} \underline{\mu}_1 - \Sigma_{12} \Sigma_{22}^{-1} \underline{\mu}_2 \\ \underline{\mu}_2 \end{pmatrix} = \begin{pmatrix} \underline{\nu}_1 \\ \underline{\nu}_2 \end{pmatrix}$$

$$\begin{aligned} \text{cov}(\underline{y}) &= E \begin{pmatrix} (\underline{y}_1 - \underline{\nu}_1)(\underline{y}_1 - \underline{\nu}_1)' & (\underline{y}_1 - \underline{\nu}_1)(\underline{y}_2 - \underline{\nu}_2)' \\ (\underline{y}_2 - \underline{\nu}_2)(\underline{y}_1 - \underline{\nu}_1)' & (\underline{y}_2 - \underline{\nu}_2)(\underline{y}_2 - \underline{\nu}_2)' \end{pmatrix} \\ &= \begin{pmatrix} \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} & 0 \\ 0 & \Sigma_{22} \end{pmatrix} = \begin{pmatrix} \Sigma_{11.2} & 0 \\ 0 & \Sigma_{22} \end{pmatrix}. \end{aligned}$$

$$\underline{Y}_1 \sim N(\underline{\mu}_1, \Sigma_{11.2})$$

$$\underline{Y}_2 \sim N(\underline{\mu}_2, \Sigma_{22}) \equiv N(\underline{\mu}_2, \Sigma_{22})$$

$\underline{X}_2$

Now, Find distⁿ of $\underline{X}_1 = \underline{Y}_1 + \Sigma_{12} \Sigma_{22}^{-1} \underline{Y}_2$ [$\underline{Y}_1, \underline{Y}_2$ are independent]

15/01/2024

CONDITIONAL DISTRIBUTIONS

We have chosen B in such a way that

$$\underline{Y}_1 = \underline{X}_1 + B \underline{X}_2$$

$$\Delta. \underline{Y}_2 = \underline{X}_2$$

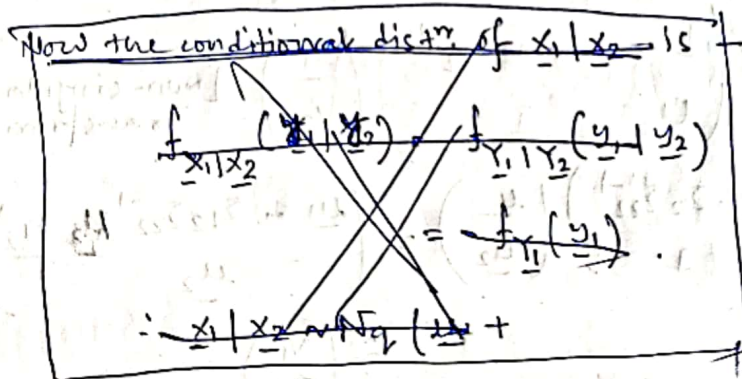
becomes independent.

and the PDF is given by -

$$f_{\underline{Y}_1, \underline{Y}_2}(\underline{y}_1, \underline{y}_2) = f_{\underline{X}_1, \underline{X}_2}(\underline{x}_1, \underline{x}_2) |J| \quad \text{with } |J| = 1$$

$$= f_{Y_1}(\underline{y}_1) \cdot f_{Y_2}(\underline{y}_2)$$

$$\text{where } \begin{cases} \underline{Y}_1 \sim N(\underline{\mu}_1 - \Sigma_{12} \Sigma_{22}^{-1} \underline{\mu}_2, \Sigma_{11.2}) \\ \underline{Y}_2 \sim N(\underline{\mu}_2, \Sigma_{22}) \end{cases}$$



Now,

$$f_{\underline{X}_1 | \underline{X}_2}(\underline{x}_1 | \underline{x}_2) = f_{Y_1}(\underline{y}_1)$$

$$\underline{Y}_1 | \underline{X}_2 \sim N(\underline{\mu}_1 - \Sigma_{12} \Sigma_{22}^{-1} \underline{x}_2, \Sigma_{11.2})$$

$$\underline{X}_1 + B \underline{X}_2 | \underline{X}_2 \sim N(\underline{\mu}_1 - \Sigma_{12} \Sigma_{22}^{-1} \underline{x}_2, \Sigma_{11.2})$$

$$\underline{X}_1 | \underline{X}_2 \sim N(\underline{\mu}_1 - B \underline{x}_2 - \Sigma_{12} \Sigma_{22}^{-1} \underline{x}_2, \Sigma_{11.2}) \equiv (\underline{\mu}_1 + \Sigma_{12} \Sigma_{22}^{-1} (\underline{x}_2 - \underline{\mu}_2), \Sigma_{11.2})$$

MULTIPLE & PARTIAL

CORRELATION COEFFICIENTS

In Regression theory,

For ~~multiple~~ simple linear regression model $Y = \beta_0 + \beta_1 X + e$

$$\hat{\beta}_1 = \frac{\text{cov}(X, Y)}{\text{var}(Y)} = \frac{\sigma_{12}}{\sigma_{22}} = \sigma_{12} \sigma_{22}^{-1}$$

distr. of

Last time, we have derived $Y_1 = X_1 + B X_2$ with $B = -\Sigma_{12} \Sigma_{22}^{-1}$

Note that, the two terms are comparable.

In the last conditional distr., the conditional mean was obtained as -

$$E(X_1 | X_2) = \mu_1 + B(X_2 - \mu_2) \\ = \mu_1 + \Sigma_{12} \Sigma_{22}^{-1} (X_2 - \mu_2)$$

So, $\Sigma_{12} \Sigma_{22}^{-1}$ can be said the matrix of regression coefficients of X_1 on X_2 . $\therefore \boxed{\beta = \Sigma_{12} \Sigma_{22}^{-1}}$

The $(i, k+1)$ th element of β is ~~$\beta_{ik, k+1, \dots, p}$~~
 $\beta_{ik, k+1, \dots, p}$.

The (i, j) th element of $\Sigma_{11.2}$ is $\sigma_{ij, k+1, \dots, p}$.
Partial Covariance

Partial Correlation coefficient is -

$$\frac{\sigma_{ij, k+1, \dots, p}}{\sqrt{\sigma_{ii, k+1, \dots, p} \cdot \sigma_{jj, k+1, \dots, p}}}$$

RESULT

16/07/2024

1. Let $\underline{x}_1, \dots, \underline{x}_N$ be N vectors, and $\underline{b}$ be any vector. Then-

$$\sum_{\alpha=1}^N (\underline{x}_\alpha - \underline{b})(\underline{x}_\alpha - \underline{b})' = \sum_{\alpha=1}^N (\underline{x}_\alpha - \bar{\underline{x}})(\underline{x}_\alpha - \bar{\underline{x}})' + N(\bar{\underline{x}} - \underline{b})(\bar{\underline{x}} - \underline{b})'$$

Derivation

$$\begin{aligned} & \sum_{\alpha=1}^N (\underline{x}_\alpha - \underline{b})(\underline{x}_\alpha - \underline{b})' \\ &= \sum_{\alpha=1}^N (\underline{x}_\alpha - \bar{\underline{x}} + \bar{\underline{x}} - \underline{b})(\underline{x}_\alpha - \bar{\underline{x}} + \bar{\underline{x}} - \underline{b})' \\ &= \sum_{\alpha=1}^N (\underline{x}_\alpha - \bar{\underline{x}})(\underline{x}_\alpha - \bar{\underline{x}})' + \sum_{\alpha=1}^N (\bar{\underline{x}} - \underline{b})(\bar{\underline{x}} - \underline{b})' \end{aligned}$$

$$+ 2 \sum_{\alpha=1}^N (\underline{x}_\alpha - \bar{\underline{x}})(\bar{\underline{x}} - \underline{b})' \rightarrow 0$$

2. Suppose, in the last result $\underline{b} = \underline{\mu}$.
Consider the quantity

$$\begin{aligned} & \sum_{\alpha=1}^N (\underline{x}_\alpha - \underline{\mu})' \Sigma^{-1} (\underline{x}_\alpha - \underline{\mu}) \\ &= \text{tr} \left[\sum_{\alpha=1}^N \Sigma^{-1} (\underline{x}_\alpha - \underline{\mu})(\underline{x}_\alpha - \underline{\mu})' \right] \\ &= \text{tr} \Sigma^{-1} \left[A + N(\bar{\underline{x}} - \underline{\mu})(\bar{\underline{x}} - \underline{\mu})' \right] \\ & \quad \text{where } A = \sum_{\alpha=1}^N \frac{(\underline{x}_\alpha - \bar{\underline{x}})(\underline{x}_\alpha - \bar{\underline{x}})'}{(\underline{x}_\alpha - \bar{\underline{x}})'(\underline{x}_\alpha - \bar{\underline{x}})} \\ &= \text{tr} \Sigma^{-1} A + N(\bar{\underline{x}} - \underline{\mu})' \Sigma^{-1} (\bar{\underline{x}} - \underline{\mu}). \end{aligned}$$

LEMMA

3. If D is p.d. of order $p \times p$, the maximum of

$$f(G) = -N \log |G| - \text{tr} G^{-1} D.$$

w.r.t. a pd matrix G exists & occurs at

$$G = \frac{1}{N} D.$$

► CHOLESKY DECOMPOSITION

If A is p.d., $\exists$ a unique lower triangular matrix T ($t_{ij} = 0 \ \forall i < j$) with positive diagonal elements. Such that $A = TT'$.

[e.g.] $T = \begin{pmatrix} a & 0 & 0 \\ b & d & 0 \\ c & e & f \end{pmatrix}$ $|T|^2 = \begin{vmatrix} a^2 & 0 & 0 \\ ab+bd & d^2 & 0 \\ ac+be+cf & de+ef & f^2 \end{vmatrix}$

$$\text{tr}(TT') = a^2 + b^2 + c^2 + d^2 + e^2 + f^2 = a^2 d^2 f^2$$

$$\log |T|^2 = \log a^2 + \log d^2 + \log f^2$$

Proof Let, $D = EE'$ (D is p.d.).

and $E' G^{-1} E = H$

and $G = EH^{-1}E'$

So, $|G| = |E| |H^{-1}| |E'|$
 $= |H^{-1}| |EE'|$

$= \frac{|D|}{|H|}$

$\text{tr}(G^{-1}D) = \text{tr}(G^{-1}EE')$
 $= \text{tr}(E'G^{-1}E) =$
 $= \text{tr}(H)$

$f(G) = -N \log |G| - \text{tr} G^{-1}D$

$= -N \log |D| + N \log |H| - \text{tr} H$

$= -N \log |D| + N \log |TT'| - \text{tr} |TT'|$ [$T = \text{chol}(H)$]

$= -N \log |D| + N \log |T|^2 - \text{tr} |T|^2$

$$= -N \log |D| + \sum_{i=1}^p N \log t_{ii}^2 = - \sum_{i=1}^p t_{ii}^2$$

$$= -N \log |D| + \sum_{i=1}^p (N \log t_{ii}^2 - t_{ii}^2)$$

$$= -N \log |D| + \sum_{i=1}^p (N \log t_{ii}^2 - t_{ii}^2)$$

$$= -N \log |D| + \sum_{i=1}^p (N \log t_{ii}^2 - t_{ii}^2)$$

$$\frac{\partial f}{\partial t_{ii}^2} = \frac{N}{t_{ii}^2} - 1 = 0 \Rightarrow N = t_{ii}^2$$

$$\frac{\partial^2 f}{\partial (t_{ii}^2)^2} = -\frac{N}{t_{ii}^4} \Rightarrow \frac{\partial^2 f}{\partial (t_{ii}^2)^2} \bigg|_{t_{ii}^2=N} = -\frac{1}{N} < 0$$

Putting, $N = t_{ii}^2$ then $H = N \cdot I_p$

then $G = E H^{-1} E'$

$$= \frac{1}{N} E E'$$

$$= \frac{1}{N} D$$

~~f(G)~~ So, at $G = \frac{1}{N} D$

$$f\left(\frac{1}{N} D\right) = -N \log \left|\frac{D}{N}\right| + N \text{tr}(D^{-1} D)$$

$$= -N \log |D| + p N \log N - N p$$

MLE

$$\underline{x}_1, \underline{x}_2, \dots, \underline{x}_N \sim N_p(\underline{\mu}, \Sigma), \quad N > p$$

$$L = \frac{1}{(2\pi)^{\frac{Np}{2}} |\Sigma|^{\frac{N}{2}}} \exp \left[-\frac{1}{2} \sum_{\alpha=1}^N (\underline{x}_\alpha - \underline{\mu})' \Sigma^{-1} (\underline{x}_\alpha - \underline{\mu}) \right]$$

$$\log L = -\frac{Np}{2} \log 2\pi - \frac{N}{2} \log |\Sigma| - \frac{1}{2} \sum_{\alpha=1}^N (\underline{x}_\alpha - \underline{\mu})' \Sigma^{-1} (\underline{x}_\alpha - \underline{\mu})$$

$$= -\frac{Np}{2} \log 2\pi - \frac{N}{2} \log |\Sigma| - \frac{1}{2} \text{tr} \Sigma^{-1} A - \frac{N}{2} (\bar{\underline{x}} - \underline{\mu})' \Sigma^{-1} (\bar{\underline{x}} - \underline{\mu})$$

(I) (II) (III)

Minimise (III) $\Sigma > 0$, so $\Sigma^{-1} > 0$, $(\bar{\underline{x}} - \underline{\mu})' \Sigma^{-1} (\bar{\underline{x}} - \underline{\mu}) \geq 0$.

Minimizing the term, we get $\hat{\underline{\mu}}_{MLE} = \bar{\underline{x}}$.

Using the lemma, $f(\Sigma) = -N \log |\Sigma| - \text{tr} \Sigma^{-1} A$

Max. occurs at $\Sigma = A/N$.

$$= \frac{1}{N} \sum_{\alpha=1}^N (\underline{x}_\alpha - \bar{\underline{x}}) (\underline{x}_\alpha - \bar{\underline{x}})'$$
$$= \hat{\Sigma}_{MLE}$$

$$E(\hat{\underline{\mu}}) = E(\bar{\underline{x}}) = E \begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \vdots \\ \bar{x}_p \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{pmatrix} = \underline{\mu}$$

Define $S = A/N-1$

$$S = \begin{pmatrix} s_{11} & s_{12} & & \\ & \ddots & \ddots & \\ & & s_{pp} \end{pmatrix}$$

$$E(S) = \Sigma$$

MLE is not unbiased, ~~but~~ S is unbiased

17/01/2024

DISTRIBUTION OF SAMPLE MEAN VECTOR

$$\underline{x} \sim N_p(\underline{\mu}, \Sigma).$$

$$f(\underline{x}_1, \dots, \underline{x}_N) = \frac{1}{(2\pi)^{Np/2} |\Sigma|^{N/2}} \exp \left[-\frac{1}{2} \sum_{\alpha=1}^N (\underline{x}_\alpha - \underline{\mu})' \Sigma^{-1} (\underline{x}_\alpha - \underline{\mu}) \right]$$

Apply the transformation

$\underline{x}^* = \underline{x} \underline{Q}$, $\underline{Q}$ is orthogonal matrix
s.t. first column elements
are $1/\sqrt{N}$ and all others
are q_{ij}

$$\underline{Q} = \begin{pmatrix} 1/\sqrt{N} & & \\ & \ddots & \\ & & 1/\sqrt{N} & q_{ij} \end{pmatrix} \quad \underline{x} = \begin{pmatrix} x_{11} & \dots & x_{1N} \\ \vdots & & \vdots \\ x_{p1} & \dots & x_{pN} \end{pmatrix}$$

Now

$$\underline{x}^* = \underline{x} \underline{Q} = \begin{bmatrix} \sqrt{N} \bar{\underline{x}} & \underline{Y} \end{bmatrix}$$

Reduced matrix.

$$\underline{x}^* \underline{x}^{*'} = \underline{x} \underline{Q} \underline{Q}' \underline{x}' = \underline{x} \underline{x}'$$

$$\underline{x}^* \underline{x}^{*'} = N \bar{\underline{x}} \bar{\underline{x}}' + \underline{Y} \underline{Y}'$$

$$\therefore \underline{Y} \underline{Y}' = \underline{x}^* \underline{x}^{*'} - N \bar{\underline{x}} \bar{\underline{x}}'$$

$$= \underline{x} \underline{x}' - N \bar{\underline{x}} \bar{\underline{x}}'$$

$$\sum_{\alpha=1}^N \underline{y}_\alpha \underline{y}_\alpha' = \sum_{\alpha=1}^N (\underline{x}_\alpha - \bar{\underline{x}}) (\underline{x}_\alpha - \bar{\underline{x}})'$$

$$= A = (N-1) S. \quad [\text{See last page notations}]$$

The likelihood is -

$$f(\underline{x}_1, \dots, \underline{x}_N) = \frac{1}{(2\pi)^{Np/2} |\Sigma|^{N/2}} \exp \left[-\frac{1}{2} \sum_{\alpha=1}^N (\underline{x}_\alpha - \underline{\mu})' \Sigma^{-1} (\underline{x}_\alpha - \underline{\mu}) \right]$$

$$= k^* \exp \left[-\frac{1}{2} + \pi \Sigma^{-1} \sum_{\alpha=1}^N (\underline{x}_\alpha - \underline{\mu})' (\underline{x}_\alpha - \underline{\mu}) \right]$$

$$= k^* \exp \left[-\frac{1}{2} \text{tr} \Sigma^{-1} \left\{ \underline{Y} \underline{Y}' + N(\underline{\bar{x}} - \underline{\mu})' (\underline{\bar{x}} - \underline{\mu}) \right\} \right]$$

$$= \frac{1}{(2\pi)^{Np/2} |\Sigma|^{N/2}} \exp \left[-\frac{1}{2} \text{tr} \Sigma^{-1} \underline{Y} \underline{Y}' \right] \exp \left[-\frac{1}{2} (\underline{\bar{x}} - \underline{\mu})' \left(\frac{\Sigma}{N} \right)^{-1} (\underline{\bar{x}} - \underline{\mu}) \right]$$

$$= \frac{1}{(2\pi)^{Np/2} |\Sigma|^{N/2}} \exp \left[-\frac{1}{2} \underline{Y}' \Sigma^{-1} \underline{Y} \right] \exp \left[-\frac{1}{2} (\underline{\bar{x}} - \underline{\mu})' \left(\frac{\Sigma}{N} \right)^{-1} (\underline{\bar{x}} - \underline{\mu}) \right]$$

$\Rightarrow \underline{\bar{x}}$ & $\underline{Y}$ are independent. — (i)

$$(ii) \underline{\bar{x}} \sim N_p(\underline{\mu}, \frac{\Sigma}{N})$$

$$(iii) \underline{Y}_i \sim N(0, \Sigma), i = 1(1)N-1$$

~~Also, $\underline{Y}$ is~~

The matrix $\underline{Y}$ follows Wishart distribution.

$$f_{\underline{Y}}(\underline{A}) = \frac{|\underline{A}|^{N-p-1/2} \exp \left[-\frac{1}{2} \text{tr} \Sigma^{-1} \underline{A} \right]}{2^{Np/2} \pi^{p(N-1)/2} |\Sigma|^{N/2} \prod_{i=1}^{N-1} \Gamma \left(\frac{N-i}{2} \right)}$$

Now, if $\underline{x} \sim N_p(\underline{\mu}, \Sigma)$

$$\underline{x}' \Sigma^{-1} \underline{x} \sim \text{Non-central } \chi^2(p, \underline{\mu}' \Sigma^{-1} \underline{\mu})$$

Let, $\underline{c}$ be a n.s. matrix s.t. $\underline{c} \Sigma \underline{c}' = \underline{I}$.

$$\underline{z} = \underline{c} \underline{x} \sim N(\underline{c} \underline{\mu}, \underline{c} \Sigma \underline{c}')$$

$$\sim N(\underline{\lambda}, \underline{I})$$

$$\underline{x}' \Sigma^{-1} \underline{x} = \underline{z}' \underline{c}'^{-1} \Sigma^{-1} \underline{c} \underline{z} = \underline{z}' (\underline{c} \Sigma \underline{c}')^{-1} \underline{z} = \underline{z}' \underline{z}$$

Then distribution of $\underline{x}' \Sigma^{-1} \underline{x} = \sum z_i^2 \sim \text{N.c. } \chi^2(p, \sum \lambda_i^2)$

$$(\bar{x} - \mu)' \left(\frac{\Sigma}{n} \right)^{-1} (\bar{x} - \mu) \sim \chi^2(p)$$

$$\sqrt{n} (\bar{x} - \mu) \sim N(0, \Sigma)$$

$$N (\bar{x} - \mu)' \Sigma^{-1} (\bar{x} - \mu) \sim \chi^2(p)$$

For unknown Σ , to test $H_0: \mu = \mu_0$, we need Hotelling's T^2 -statistic.

23/07/2024

Theorem $\rightarrow$ If A & B are two real symmetric matrix st. B is pd, then there exists a n.s. matrix P s.t. $P'AP = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$, $P'BP = I_n$ where $\lambda_1, \dots, \lambda_n$ are the ~~roots~~ of the equation $|A - \lambda B| = 0$.

Proof $\rightarrow$ If B is pd., $\exists$ a n.s. matrix Q s.t. $Q'BQ = I$.
Consider $Q'AQ$, real symmetric.
So by spectral decomposition theorem, $\exists$ an orthogonal matrix H s.t.

$$H'(Q'AQ)H = D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$$

where $\lambda_1, \dots, \lambda_n$ are the roots of $|Q'AQ - \lambda I| = 0$.

$$\text{Let } P = QH \text{ (P is n.s.)}$$

$$\Rightarrow P'AP = D = \text{diag}(\lambda_1, \dots, \lambda_n)$$

$$[Q \text{ is n.s.} \Rightarrow P \text{ is also n.s.}]$$

$$Q'BQ = I \text{ or } H'QBQ^{-1} = H'H = I$$

or $P'BP = I$ (*)

Wishart distribution

$$x_1, x_2, \dots, x_n \sim N(\mu, \Sigma)$$

$$\hat{\mu} = \bar{x}, \hat{\Sigma} = S = \frac{A}{n-1} = \frac{A}{n}$$

$$A = \sum_{\alpha=1}^n (x_\alpha - \bar{x})(x_\alpha - \bar{x})', \quad n = N-1$$

(P.D. matrix) $(N-1)S = A$

$$S = \begin{pmatrix} s_{11} & s_{12} & \dots & s_{1p} \\ s_{21} & & & \\ \vdots & & & \\ s_{p1} & s_{p2} & \dots & s_{pp} \end{pmatrix} \rightarrow \frac{p(p+1)}{2} \text{ unique elements.}$$

Let, $x_1, \dots, x_N$ ($N > p$) are ind. of each other and they follow $N(\underline{\mu}, \Sigma)$ distribution. If a random pd matrix A , then the PDF -

$$f(A) = \frac{|A|^{n-p-1/2} \exp\left(-\frac{1}{2} \text{tr} \Sigma^{-1} A\right)}{2^{np/2} |\Sigma|^{n/2} \pi^{p(p-1)/4} \prod_{i=0}^{p-1} \Gamma\left(\frac{n-i}{2}\right)}$$

Then A is said to have a Wishart distⁿ with parameter Σ and n .

$$\text{i.e. } \boxed{A \sim W(\Sigma, n)}$$

Theorem: If $\gamma_\alpha \sim N_p(0, \Sigma)$, $\alpha = 1, \dots, n$ and are independently distributed then the density of $A = \sum_{\alpha=1}^n \gamma_\alpha \gamma_\alpha' \sim W(\Sigma, n)$ for pd A & $n > p$. [no density if $n < p$]

$$\begin{aligned} \star \quad |B'AB - \lambda I| &= 0 \\ \Rightarrow |B'AB - \lambda B'B B| &= 0 \\ \Rightarrow |B'| |A - \lambda B| |B| &= 0 \\ \Rightarrow |B B'| |A - \lambda B| &= 0 \\ \Rightarrow |A - \lambda B| &= 0 \end{aligned}$$

Characteristic Function

$$\gamma_\alpha \stackrel{w}{\sim} N(0, \Sigma), \alpha = 1(1)n$$

$$A = \sum_{\alpha=1}^n \gamma_\alpha \gamma_\alpha'$$

$$\begin{aligned} \text{characteristic function!! } \phi(t) &= E[\exp(i \text{tr} A t)] \\ &= E\left[\exp\left(i \text{tr} \sum_{\alpha=1}^n \gamma_\alpha \gamma_\alpha' t\right)\right] \\ &= E\left[\exp\left(i \sum_{\alpha=1}^n \gamma_\alpha' t \gamma_\alpha\right)\right] \end{aligned}$$

$$= \prod_{\alpha=1}^n E [\exp(i \underline{y}'_\alpha \oplus \underline{y}_\alpha)]$$

$$= \left\{ E [\exp(i \underline{y}' \oplus \underline{y})] \right\}^n$$

Consider the quantity,

$$E [\exp(i \underline{y}' \oplus \underline{y})]$$

Consider Σ^{-1} and Θ real, $\exists$ a real

n.s. matrix P s.t. $P' \Sigma^{-1} P = I$ and

$$P' \oplus P = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_p) = D$$

where $\lambda_1, \dots, \lambda_p$ are the characteristic roots

$$\text{of } |\oplus - \lambda I| = 0$$

$$\text{Now, } \underline{z} \sim N(0, I)$$

Consider a linear transformation, $\underline{y} = P \underline{z}$, z_j 's are independently distributed.

$$E [\exp(i \underline{y}' \oplus \underline{y})]$$

$$= E [\exp(i \underline{z}' P' \oplus P \underline{z})]$$

$$= E [\exp(i \sum_{j=1}^p \lambda_j z_j^2)] \quad [z_j \sim N(0, 1)]$$

$$= \prod_{j=1}^p (1 - 2i\lambda_j)^{-1/2}$$

$$= |I - 2iD|^{-1/2}$$

$$= |P' \Sigma^{-1} P - 2i P' \oplus P|^{-1/2}$$

$$= [|P'| | \Sigma^{-1} - 2i \oplus | |P|]^{-1/2}$$

$$= [|P|^2 | \Sigma^{-1} - 2i \oplus |]^{-1/2}$$

$$= \frac{| \Sigma^{-1} - 2i \oplus |^{-1/2}}{|P'|^2 |P|^{1/2}} = \frac{| \Sigma^{-1} - 2i \oplus |^{-1/2}}{| \Sigma^{-1} |^{1/2}}$$

$$= |I - 2i \oplus \Sigma|^{-1/2}$$

$$\phi(\oplus) = |I - 2i \oplus \Sigma|^{-1/2}$$

► Additive Property

If $A_j (j=1, \dots, r)$ i.i.d $W(\Sigma, n_j)$.
 then, $A = \sum_{j=1}^r A_j \sim W(\Sigma, \sum_{j=1}^r n_j)$.

Proof $\phi_{A_j}(\Theta) = |I - 2i\Theta\Sigma|^{-n_j/2}$

$$\begin{aligned}\phi_A(\Theta) &= \phi_{\sum_{j=1}^r A_j}(\Theta) = \prod_{j=1}^r \phi_{A_j}(\Theta) \\ &= |I - 2i\Theta\Sigma|^{-\sum_{j=1}^r n_j/2} \\ &= |I - 2i\Theta\Sigma|^{-n/2} \dots |I - 2i\Theta\Sigma|^{-n_j/2}\end{aligned}$$

Hence, By uniqueness theorem,

$A \sim W(\Sigma, \sum_{j=1}^r n_j)$. [This is also called Reproductive Property]

24/01/2024

► DISTRIBUTION OF LINEAR TRANSFORMATION

If $A \sim W(\Sigma, n)$, $C: p \times p$ n.s. matrix then for
 $B = C'AC \sim W(C'\Sigma C, n)$ where C is s.t. $\underline{Z}_\alpha = C' \underline{Y}_\alpha$.

$A = \Sigma \underline{Y}_\alpha \underline{Y}_\alpha'$
 $\underline{Y}_\alpha \sim N_p(0, \Sigma) \quad \forall \alpha = 1, \dots, n$
 $\underline{Z}_\alpha = C' \underline{Y}_\alpha$

$B = C'AC = C'(\Sigma \underline{Y}_\alpha \underline{Y}_\alpha')C = \Sigma (C' \underline{Y}_\alpha)(C' \underline{Y}_\alpha)'$
 $= \Sigma \underline{Z}_\alpha \underline{Z}_\alpha'$

$\underline{Y}_\alpha \sim N(0, \Sigma) \Rightarrow \underline{Z}_\alpha \sim N(0, C'\Sigma C)$

$\therefore \Sigma \underline{Z}_\alpha \underline{Z}_\alpha' \sim W(C'\Sigma C, n)$

► MARGINAL DISTRIBUTION

$$A = \sum \underline{y}_\alpha \underline{y}_\alpha', \quad \underline{y}_\alpha \sim N(0, \Sigma), \quad \underline{y}_\alpha = \begin{pmatrix} y_{\alpha(1)} \\ y_{\alpha(2)} \end{pmatrix}$$

where, $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

So, $A_{11} \sim W(\Sigma_{11}, n)$

~~Let~~ $\begin{cases} \underline{y}_\alpha \sim N(\underline{\mu}, \Sigma) \\ \& A_{11} = \sum_{\alpha=1}^n \underline{y}_\alpha^{(1)} \underline{y}_\alpha^{(1)'} \sim W(\Sigma_{11}, n) \end{cases}$

In general,

$$\underline{y}_\alpha = \begin{bmatrix} y_{\alpha(1)} \\ \vdots \\ y_{\alpha(p)} \end{bmatrix}$$

$$A = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1p} \\ \vdots & A_{22} & \dots & \vdots \\ A_{p1} & \dots & A_{pp} \end{bmatrix} \quad \Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} & \dots & \Sigma_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ \Sigma_{p1} & \dots & \dots & \Sigma_{pp} \end{bmatrix}$$

g) $\Sigma_{ij} = 0 \forall i \neq j$ and $A \sim W(\Sigma, n)$ then

$A_{11}, A_{22}, \dots, A_{pp}$ are independently distributed with

$A_{jj} \sim W(\Sigma_{jj}, n)$

Also,

g) $\Sigma_{ij} = 0 \Rightarrow \underline{y}_\alpha^{(1)}, \underline{y}_\alpha^{(2)}, \dots, \underline{y}_\alpha^{(p)} \quad \alpha = 1, \dots, n$

are independently distributed.

$A_{11} = \sum \underline{y}_\alpha^{(1)} \underline{y}_\alpha^{(1)'}, \dots, A_{pp} = \sum \underline{y}_\alpha^{(p)} \underline{y}_\alpha^{(p)'}$

are also independent, $A_{jj} \sim W_{p_j}(\Sigma_{jj}, n) \quad j=1, \dots, p$

Consider the quantity,

$$A_{22.1} = A_{22} - A_{21} A_{11}^{-1} A_{12}$$

$$\underline{y}_\alpha \sim N(0, \Sigma)$$

$$\rightarrow = \begin{pmatrix} y_\alpha^{(1)} \\ y_\alpha^{(2)} \end{pmatrix} \quad A = \sum y_\alpha y_\alpha' = Y^T Y \quad \text{where } Y = \begin{pmatrix} y_1' \\ \vdots \\ y_n' \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$\text{Now, } A_{22.1} = A_{22} - A_{21} A_{11}^{-1} A_{12}$$

$$= y_2' y_2 - y_2' y_1 A_{11}^{-1} y_1' y_2$$

$$= y_2' [I - y_1 A_{11}^{-1} y_1'] y_2$$

$$= y_2' P y_2$$

$$P = I - y_1 A_{11}^{-1} y_1'$$

Symmetric, idempotent

$$\text{rank}(P) = n - q$$

$$y_1 P = 0$$

$$\text{Define, } y_{2.1} = y_2 - y_1 \Sigma_{11}^{-1} \Sigma_{12} \quad [\equiv y_2 | y_1 = y_1]$$

$$\text{Cov}(y_{2.1}, y_1) = (\Sigma_{12} - \Sigma_{11} \Sigma_{11}^{-1} \Sigma_{12}) = 0$$

$$y_{2.1}' P y_{2.1} = y_2' y_2 - y_2' y_1 A_{11}^{-1} y_1' y_2 = y_2' P y_2 = A_{22.1}$$

[Derive]

$$\text{Therefore } y_{2.1} \sim N(\mu_{2.1}, \Sigma_{11.2})$$

$$\& \sum_{\alpha=1}^n y_{2.1} y_{2.1}' \sim W(\Sigma_{11.2}, n)$$

GENERALIZED VARIANCE

Multivariate analogue of σ^2 is Σ . ($|\Sigma|$ also).

Sample version: $|S| = \frac{1}{N-1} \sum_{\alpha=1}^N (\underline{x}_{\alpha} - \bar{\underline{x}})(\underline{x}_{\alpha} - \bar{\underline{x}})'$

Result // Given a random sample $x_1, \dots, x_N$ from $N_p(\mu, \Sigma)$, the distribution of $\frac{|S|}{(N-1)}$ times the product of independent factors, where the i th factor follows χ^2_{n-i} distribution.

i.e. $f(|S|) \sim \frac{|S|}{Np} f(\chi^2_1) \dots f(\chi^2_{N-p+1})$

ASYMPTOTIC DISTRIBUTION

$$\sqrt{N} \left(\frac{|S|}{|\Sigma|} - 1 \right) \sim N(0, 2p)$$

$$E(|S|) = \frac{|\Sigma|}{Np} \prod_{i=1}^p E(\chi^2_{N-i})$$

$$= \frac{|\Sigma|}{Np} \prod_{i=1}^p (N-i)$$

COCHRAN'S THEOREM

Suppose $\underline{y}_1, \dots, \underline{y}_N \stackrel{iid}{\sim} N(0, \Sigma)$

Rank $(C_i) = m$, suppose $C_i = ((C_{\alpha\beta}^i))$

$$\sum_{i=1}^m C_i = I_N$$

$$B_i = \sum_{\alpha=1}^N \sum_{\beta=1}^N C_{\alpha\beta}^i \underline{y}_{\alpha} \underline{y}_{\beta}'$$

$$\sum_{i=1}^m B_i = \sum_{\alpha=1}^N \underline{y}_{\alpha} \underline{y}_{\alpha}'$$

A necessary and sufficient condition that $B_1, \dots, B_m$ are independently distributed with $B_i \sim W(\Sigma, n_i)$.

$$\sum_{i=1}^m n_i = N$$

INVERTED WISHART DISTRIBUTION

$$A \sim W(\Sigma, n)$$

$$B = A^{-1} \sim f(B) = \frac{|\Sigma^{-1}|^{n/2} |B|^{-\frac{n+p-1}{2}}}{2^{np/2} |\Sigma|^{(n-p)/2} \pi^{p(p-1)/4} \prod_{i=0}^{p-1} \Gamma\left(\frac{n-i}{2}\right)}$$

[Derive]

$$\exp\left[-\frac{1}{2} \text{tr}(\Sigma^{-1} B^{-1})\right]$$

► $y_\alpha \sim N(0, \Sigma)$

$$\Sigma y_\alpha y_\alpha' \sim W(\Sigma, n)$$

$x_\alpha \sim N(\mu, \Sigma)$

$$A = \Sigma x_\alpha x_\alpha' \sim \text{Non-central Wishart}(\Sigma, n, \Sigma^{-1} \mu \mu')$$

► $A \sim W(\Sigma, n)$

$$E(A) = n\Sigma \quad \& \quad E(A^{-1}) = \frac{\Sigma^{-1}}{n-p-1}$$

GENERALISED T^2 STATISTIC

29/01/2024.

Univariate $H_0: \mu = \mu_0, x_1, \dots, x_n \sim N(\mu, \sigma^2)$

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t_{n-1}$$

Multivariate $H_0: \underline{\mu} = \underline{\mu}_0$

$$\underline{x}_1, \dots, \underline{x}_n \sim N_p(\underline{\mu}, \underline{\Sigma})$$

$$T^2 = N (\bar{\underline{x}} - \underline{\mu})' \underline{\Sigma}^{-1} (\bar{\underline{x}} - \underline{\mu})$$

~~Hotelling~~ T^2 statistic

Sample version: (6)

Result For any $|B| \neq 0$.

$$\begin{vmatrix} B & C \\ D & E \end{vmatrix} = \begin{vmatrix} B & C \\ D - EB^{-1}C & E - CB^{-1}C \end{vmatrix}$$

$$= \begin{vmatrix} B & 0 \\ D & E - DB^{-1}C \end{vmatrix}$$

$$= |B| |E - DB^{-1}C|$$

e.g.

$$\begin{vmatrix} A & \sqrt{N}(\bar{\underline{x}} - \underline{\mu}) \\ \sqrt{N}(\bar{\underline{x}} - \underline{\mu})' & I \end{vmatrix}$$

$$= \begin{vmatrix} A & \sqrt{N}(\bar{\underline{x}} - \underline{\mu})' \\ -\sqrt{N}(\bar{\underline{x}} - \underline{\mu}) & I \end{vmatrix}$$

$$= \begin{vmatrix} A & \sqrt{N}(\bar{\underline{x}} - \underline{\mu})' \\ -\sqrt{N}(\bar{\underline{x}} - \underline{\mu}) & I \end{vmatrix} \begin{vmatrix} I & -\sqrt{N}A^{-1}(\bar{\underline{x}} - \underline{\mu})' \\ 0 & I \end{vmatrix}$$

$$= \begin{vmatrix} A & 0 \\ -\sqrt{N}(\bar{\underline{x}} - \underline{\mu}) & I + N(\bar{\underline{x}} - \underline{\mu})A(\bar{\underline{x}} - \underline{\mu})' \end{vmatrix}$$

LIKELIHOOD RATIO TEST

$$\underline{X} \sim N_p(\underline{\mu}, \underline{\Sigma})$$

$\underline{x}_1, \dots, \underline{x}_N \rightarrow \text{sample.}$

$$L = L(\underline{\mu}, \underline{\Sigma}) = \frac{1}{(2\pi)^{Np/2} |\underline{\Sigma}|^{N/2}} \exp \left[-\frac{1}{2} \sum_{\alpha=1}^N (\underline{x}_{\alpha} - \underline{\mu})' \underline{\Sigma}^{-1} (\underline{x}_{\alpha} - \underline{\mu}) \right]$$

The LR test statistic is -

$$\lambda = \frac{\text{Max}_{H_0} L}{\text{Max}_{\Omega} L}$$

$$\hat{\Omega}_{H_0} = (\hat{\underline{\mu}}, \hat{\underline{\Sigma}}) = (\underline{\mu}_0, \hat{\underline{\Sigma}}_w)$$

$$\hat{\Omega}_{\Omega} = (\hat{\underline{\mu}}_{\Omega}, \hat{\underline{\Sigma}}_{\Omega}) = (\bar{\underline{x}}, \frac{\underline{A}}{N})$$

$$\text{where } \begin{cases} \hat{\underline{\Sigma}}_{\Omega} = \frac{1}{N} \sum_{\alpha=1}^N (\bar{\underline{x}}_{\alpha} - \bar{\underline{x}}) (\bar{\underline{x}}_{\alpha} - \bar{\underline{x}})' \\ \hat{\underline{\Sigma}}_w = \frac{1}{N} \sum_{\alpha=1}^N (\underline{x}_{\alpha} - \underline{\mu}_0) (\underline{x}_{\alpha} - \underline{\mu}_0)' \end{cases}$$

$$\text{Max}_{H_0} L = \frac{1}{(2\pi)^{Np/2} |\hat{\underline{\Sigma}}_w|^{N/2}} \exp \left[-\frac{1}{2} \sum_{\alpha=1}^N (\underline{x}_{\alpha} - \underline{\mu}_0)' \hat{\underline{\Sigma}}_w^{-1} (\underline{x}_{\alpha} - \underline{\mu}_0) \right]$$

$$= \frac{1}{(2\pi)^{Np/2} |\hat{\underline{\Sigma}}_w|^{N/2}} \exp \left[-\frac{1}{2} \text{tr} \hat{\underline{\Sigma}}_w^{-1} \sum_{\alpha=1}^N \left\{ (\underline{x}_{\alpha} - \underline{\mu}_0) (\underline{x}_{\alpha} - \underline{\mu}_0)' \right\} \right]$$

$$= \frac{1}{(2\pi)^{\frac{Np}{2}} |\hat{\underline{\Sigma}}_w|^{N/2}} \exp \left[-\frac{N}{2} \text{tr} \hat{\underline{\Sigma}}_w^{-1} \underline{\Sigma}_w \right]$$

$$= \frac{1}{(2\pi)^{Np/2} |\hat{\underline{\Sigma}}_w|^{N/2}} \exp \left[-\frac{Np}{2} \right]$$

We know, $A = \sum_{\alpha=1}^N (\underline{x}_{\alpha} - \bar{x})(\underline{x}_{\alpha} - \bar{x})'$

$$A_0 = \sum_{\alpha=1}^N (\underline{x}_{\alpha} - \underline{\mu}_0)(\underline{x}_{\alpha} - \underline{\mu}_0)'$$

$$= A + N(\bar{x} - \underline{\mu}_0)(\bar{x} - \underline{\mu}_0)' \quad \text{--- (**)}$$

Now,

$$\text{Max}_{\Sigma} L = \frac{1}{(2\pi)^{Np/2} |\hat{\Sigma}_{\Sigma}|^{N/2}} \exp \left[-\frac{1}{2} \sum_{\alpha=1}^N (\underline{x}_{\alpha} - \bar{x}) \hat{\Sigma}_{\Sigma}^{-1} (\underline{x}_{\alpha} - \bar{x})' \right]$$

$$= \frac{1}{(2\pi)^{Np/2} |\hat{\Sigma}_{\Sigma}|^{N/2}} \exp \left[-\frac{1}{2} \text{tr} \left(\hat{\Sigma}_{\Sigma}^{-1} \sum_{\alpha=1}^N \{ (\underline{x}_{\alpha} - \bar{x})(\underline{x}_{\alpha} - \bar{x})' \} \right) \right]$$

$$= \frac{1}{(2\pi)^{Np/2} |\hat{\Sigma}_{\Sigma}|^{N/2}} \exp \left[-\frac{N}{2} \text{tr} \left(\hat{\Sigma}_{\Sigma}^{-1} \hat{\Sigma}_{\Sigma} \right) \right]$$

$$= \frac{1}{(2\pi)^{Np/2} |\hat{\Sigma}_{\Sigma}|^{N/2}} \exp(-Np/2)$$

$$\therefore \lambda = \frac{|\hat{\Sigma}_{\Sigma}|^{N/2}}{|\hat{\Sigma}_W|^{N/2}} \quad \text{or, } \lambda^{2/N} = \frac{|\hat{\Sigma}_{\Sigma}|}{|\hat{\Sigma}_W|}$$

$$= \frac{\left| \sum_{\alpha=1}^N (\underline{x}_{\alpha} - \bar{x})(\underline{x}_{\alpha} - \bar{x})' \right|}{\left| \sum_{\alpha=1}^N (\underline{x}_{\alpha} - \underline{\mu}_0)(\underline{x}_{\alpha} - \underline{\mu}_0)' \right|}$$

$$= \frac{|A|}{|A|}$$

$$|A + N(\bar{x} - \underline{\mu}_0)(\bar{x} - \underline{\mu}_0)'|$$

$$= \frac{|A|}{|A|} \quad \text{[from **]}$$

$$= \frac{|A| |I + N(\bar{x} - \underline{\mu}_0)(\bar{x} - \underline{\mu}_0)'|}{|A|}$$

$$= \frac{|A| [1 + N(\bar{x} - \underline{\mu}_0)' A^{-1} (\bar{x} - \underline{\mu}_0)]}{|A|}$$

$$\lambda = \frac{2/N}{1 + \frac{T^2}{N-1}}$$

Reject H_0 , if $\lambda \leq \lambda_0$.

$$\begin{aligned} T^2 &= N(\bar{X} - \mu_0)' S^{-1} (\bar{X} - \mu_0) \\ &= N(N-1) (\bar{X} - \mu_0)' A^{-1} (\bar{X} - \mu_0) \end{aligned}$$

λ_0 is chosen s.t. $P(\lambda \leq \lambda_0) = \alpha$ when H_0 is true.

$$\lambda \leq \lambda_0 \Rightarrow \frac{N}{1 + \frac{T^2}{N-1}} \leq \lambda_0$$

$$\Rightarrow \lambda^{-2/N} - 1 \geq \lambda_0^{-2/N} - 1$$

$$\Rightarrow (N-1) (\lambda^{-2/N} - 1) \geq (N-1) (\lambda_0^{-2/N} - 1)$$

$$\Rightarrow (N-1) \left(1 + \frac{T^2}{N-1} - 1\right) \geq (N-1) \left(1 + \frac{T_0^2}{N-1} - 1\right)$$

$$\Rightarrow T^2 \geq T_0^2(\alpha), \text{ where } T_0^2(\alpha) \text{ is a const. to be determined with size condition.}$$

$$\text{Now, } T_0^2 = (N-1) (\lambda_0^{-2/N} - 1)$$

Consider Σ known then $(\bar{X} - \mu_0)' S^{-1} (\bar{X} - \mu_0) \sim \chi_p^2$.

DISTRIBUTION OF T^2

Let, $\underline{Y} \sim N_p(\underline{\mu}, \Sigma)$ and $A = nS \sim W_p(n, \Sigma)$.

$\underline{Y}$ & S are independent.

$$\text{If } T^2 = \underline{Y}' S^{-1} \underline{Y}, \text{ then } \frac{T^2}{n} \sim F(p, n-p+1, \delta)$$

non-centrality parameter.

$$\text{Proof} \rightarrow \begin{bmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{bmatrix}$$

$$\Rightarrow \begin{pmatrix} a^2 + b^2 & ac + bd \\ ac + bd & c^2 + d^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{cases} a^2 + b^2 = 1 = c^2 + d^2 \\ ac + bd = 0 \end{cases}$$

2nd original setup

$$\begin{bmatrix} T^2 \\ Y = \sqrt{n}(\bar{X} - \underline{\mu}_0) \end{bmatrix}$$

Let, D be a non-singular matrix st. $D \Sigma D' = I$.
 Define, $\underline{Y}^* = D\underline{Y}$, $S^* = DSD'$, $\underline{Y}^* = D\underline{Y}$, $\underline{Z}^* = D\underline{Z}$.

[Lemma] For any $p \times p$ n.s. matrix C and H and any vector $\underline{u}$, $\underline{u}' H^{-1} \underline{u} = (C\underline{u})' (CHC')^{-1} (C\underline{u})$

$$T^2 = \underline{Y}^{*'} S^{*-1} \underline{Y}^*$$

$$\begin{aligned} &= (D\underline{Y})' (DSD')^{-1} (D\underline{Y}) \\ &= \underline{Y}' D D^{-1} S D^{-1} D \underline{Y} \\ &= \underline{Y}' S \underline{Y} \end{aligned}$$

Now, $\underline{Y}^* \sim N_p(\underline{y}^*, D \Sigma D' = I)$

and $\underline{Y}^*$ and nS^* are independently distributed as -

$$\sum_{i=1}^n \underline{Z}_i^* \underline{Z}_i^{*'} = \sum_{i=1}^n (D \underline{Z}_i) (D \underline{Z}_i)'$$

each $\underline{Z}_i^* \sim N(0, I)$.

Define an orthogonal matrix $Q_{p \times p}$ such that first row is $q_{1i} = \frac{Y_i^*}{\sqrt{Y^{*'} Y^*}}$, $i = 1, \dots, p$.
 and remaining rows can be chosen arbitrarily.

[Lemma: $A_{n \times m}$ ($n > m$) s.t. $A'A = I_m$ then $\exists$ a matrix B , s.t. (A, B) is orthogonal]

Since, Q depends on y^* (random), it is a random matrix.

Let, $U = QY^*$, $B = Q(ns)Q'$.

$U = (U_i)$ $U_1 = \sum_i q_{1i} y_i^* = \sqrt{y^{*'} y^*}$

$U_j = \sum_i q_{ji} y_i^* = \sqrt{y^{*'} y^*} \sum_i q_{ji} q_{1i} = 0$.

$\left[\frac{T^2}{n} = y^* (Q'Q)^{-1} S^{-1} (Q'Q) y^* \right]$
 $= y^* S^{-1} y^* \cdot$

$\frac{T^2}{n} = U' B^{-1} U = (U_1, 0, \dots, 0) \begin{pmatrix} b^{11} & b^{12} & \dots & b^{1p} \\ \vdots & \vdots & \ddots & \vdots \\ b^{p1} & b^{p2} & \dots & b^{pp} \end{pmatrix} \begin{pmatrix} U_1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$
 $= U_1^2 \cdot b^{11}$

$[B^{-1} = (b^{ij})]$

$B = \begin{bmatrix} b_{11} & b_{c1} \\ b_{c1} & B_{22} \end{bmatrix} \Rightarrow \frac{1}{b^{11}} = b_{11} - b_{c1}' B_{22}^{-1} b_{c1}$
 $= b_{11.23 \dots p}$

[Recall: $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{12} & A_{22} \end{pmatrix}$.

Using that, $A_{11.2} = A_{11} - A_{12} A_{22}^{-1} A_{21} \sim W_p(n-p+1, \Sigma_{11.2})$

$b_{11.23 \dots p} \sim \chi^2(n-p+1)$

$y^{*'} y^* \sim \text{non-central } \chi^2$

$\frac{T^2}{n} = \frac{U_1^2}{b^{11}} = \frac{y^{*'} y^*}{b_{11.23 \dots p}} \sim \frac{\text{non-central } \chi^2_p}{\chi^2(n-p+1)}$

$\frac{\frac{n-p+1}{p} \cdot \frac{T^2}{n}}{\frac{N-p}{p} \cdot \frac{T^2}{N-1}} \sim \text{non-central } F(p, n-p+1, \gamma' \Sigma^{-1} \gamma)$
 $\sim \text{n.c.F. } (p, N-p, \gamma' \Sigma^{-1} \gamma)$

T^2 distribution with $n (= N-1)$ d.f.

of $H_0: \underline{\mu} = \underline{\mu}_0$, $\underline{Y} = \sqrt{N}(\bar{\underline{x}} - \underline{\mu}_0)$.

$E(\underline{Y}) = \underline{0}$

① One-sample mean Problem

$X_1, \dots, X_N \stackrel{i.i.d}{\sim} N(\underline{\mu}, \Sigma)$, $\lambda = \frac{1}{1 + \frac{T^2}{N-1}}$

W: $\lambda \leq \lambda_0 \rightarrow T^2 \geq T_0^2(\alpha)$

$T_0^2 = (N-1) (\lambda_0^{-2/N} - 1)$

So, $T_0^2 = \frac{(N-1)p}{N-p} F_{p, N-p}(\alpha)$

$= T_{p, N-1}^2(\alpha)$

$= T_{p, n}^2(\alpha)$

CONFIDENCE REGION

$\underline{\mu}$ s.t., $N(\bar{\underline{x}} - \underline{\mu})' S^{-1} (\bar{\underline{x}} - \underline{\mu}) \leq T_{p, N-1}^2(\alpha)$

Lemma: If $B_{p \times p}$ is a n.s. matrix & $\underline{v}$: $p \times 1$ vector, then $\underline{v}' B^{-1} \underline{v}$ is the non-zero root of $|\underline{v} \underline{v}' - \lambda B| = 0$

Proof: Let, λ_1 is a non-zero root of $|\underline{v} \underline{v}' - \lambda B| = 0$ with characteristic vector $\underline{\beta}$ satisfying $\underline{v} \underline{v}' \underline{\beta} = \lambda \underline{\beta}$. Since $\lambda \neq 0$, $\underline{v}' \underline{\beta} \neq 0$, we have $\underline{v}' B^{-1} \cdot \underline{v} \underline{v}' \underline{\beta} = \underline{v}' \underline{\beta}$

or, $(\underline{v}' B^{-1} \underline{v}) \cdot (\underline{v}' \underline{\beta}) = \lambda_1 (\underline{v}' \underline{\beta})$

Consider, $\underline{V} = \sqrt{N} (\bar{\underline{x}} - \underline{\mu}_0)$, $B = A$.

T^2 statistics $\rightarrow$ computed from $\bar{\underline{x}}$ & A .

$A^{-1} (\bar{\underline{x}} - \underline{\mu}_0) = \underline{b}$ is a solution of $A \underline{b} = \bar{\underline{x}} - \underline{\mu}_0$.

$$= N (\bar{\underline{x}} - \underline{\mu}_0) \underline{b}$$

$\frac{T^2}{N-1}$ is the non-zero root of $|\underline{N} (\bar{\underline{x}} - \underline{\mu}_0) (\bar{\underline{x}} - \underline{\mu}_0)' - \lambda A| = 0$.

① TWO-SAMPLE TEST

$$\begin{aligned} & \underline{x}_1^{(1)}, \underline{x}_2^{(1)}, \dots, \underline{x}_{N_1}^{(1)} \sim N(\underline{\mu}^{(1)}, \Sigma) \\ & \underline{x}_1^{(2)}, \underline{x}_2^{(2)}, \dots, \underline{x}_{N_2}^{(2)} \sim N(\underline{\mu}^{(2)}, \Sigma) \\ & \bar{\underline{x}}^{(1)}, \bar{\underline{x}}^{(2)} \\ & S_1, S_2 \end{aligned}$$

$$H_0: \underline{\mu}^{(1)} = \underline{\mu}^{(2)}$$

Pooled Covariance matrix

$$S = \frac{(N_1 - 1)S_1 + (N_2 - 1)S_2}{N_1 + N_2 - 2}$$

$$= \frac{n_1 S_1 + n_2 S_2}{n_1 + n_2}$$

$$S_1 = \frac{1}{N_1 - 1} \sum_{k=1}^{N_1} (\underline{x}_k^{(1)} - \bar{\underline{x}}^{(1)}) (\underline{x}_k^{(1)} - \bar{\underline{x}}^{(1)})'$$

$$S_2 = \frac{1}{N_2 - 1} \sum_{k=1}^{N_2} (\underline{x}_k^{(2)} - \bar{\underline{x}}^{(2)}) (\underline{x}_k^{(2)} - \bar{\underline{x}}^{(2)})'$$

$$V(\bar{\underline{x}}^{(1)} - \bar{\underline{x}}^{(2)}) = \frac{\Sigma}{N_1} + \frac{\Sigma}{N_2} = \frac{N_1 + N_2}{N_1 N_2} \Sigma$$

$$\Rightarrow V\left(\sqrt{\frac{N_1 N_2}{N_1 + N_2}} (\bar{\underline{x}}^{(1)} - \bar{\underline{x}}^{(2)})\right) = \Sigma$$

Therefore, under H_0 ,

$$\sqrt{\frac{N_1 N_2}{N_1 + N_2}} (\bar{\underline{x}}^{(1)} - \bar{\underline{x}}^{(2)}) \sim N(\underline{0}, \Sigma)$$

$$(N_1 + N_2 - 2) S \sim \sum_{\alpha=1}^{N_1 + N_2 - 2} \bar{z}_\alpha \bar{z}_\alpha' \text{ where } \bar{z}_\alpha \sim N(0, \Sigma)$$

$$T^2 = \frac{N_1 N_2}{N_1 + N_2} (\bar{x}^{(1)} - \bar{x}^{(2)})' S^{-1} (\bar{x}^{(1)} - \bar{x}^{(2)})$$

$$\sim T_{N_1 + N_2 - 2}^2 \text{ (under } H_0 \text{)}.$$

So, under H_0 , $\frac{N_1 + N_2 - p - 1}{p} \cdot \frac{T^2}{N_1 + N_2 - 2} \sim F(p, N_1 + N_2 - p - 1)$
(α).

Confidence Region

Set of vectors 'm' s.t.

$$(\bar{x}^{(1)} - \bar{x}^{(2)} - m)' S^{-1} (\bar{x}^{(1)} - \bar{x}^{(2)} - m) \leq \frac{N_1 + N_2}{N_1 N_2} T_{p, N_1 + N_2 - 2}^2(\alpha).$$

③ Problem of symmetry
(Symmetry property of T^2)

$$H_0: \mu_1 = \mu_2 = \dots = \mu_p$$

$$\text{alternatively, } H_0: \mu_1 - \mu_2 = 0,$$

$$\mu_2 - \mu_3 = 0,$$

$$\vdots$$

$$\mu_{p-1} - \mu_p = 0.$$

(p-1) linearly independent contrasts.

$$\text{Or, } H_0: C\mu = 0$$

where: C is a $(p-1) \times p$ matrix

$$\text{ \& } \underline{\mu} = (\mu_1, \dots, \mu_p)'$$

$$\text{Now, } \underline{y}_\alpha = C \underline{x}_\alpha \sim N(C\mu, C\Sigma C')$$

if under $H_0: \mu = 0$

$$T_x^2 = N(\bar{x} - 0)' S^{-1} (\bar{x} - 0) = N\bar{x}' S^{-1} \bar{x}$$

$$T_y^2 = N(C\bar{x})' (C S C')^{-1} (C\bar{x}) = N\bar{x}' C' C^{-1} S C^{-1} C \bar{x} = N\bar{x}' S^{-1} \bar{x}$$

So, basically, T^2 is invariant of a non-singular linear transformation.

Also, T^2 is independent of c .

e.g. >

$$H_0: \mu_1 - \mu_2 + \mu_3 = \mu_4.$$

$$\equiv (1 \ -1 \ 1 \ -1) \begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \end{pmatrix} = 0$$

$$H_0: C\mu = 0.$$

1) Mahalanobis Distance

$$\Delta^2 = (\underline{\mu}_1 - \underline{\mu}_2)' \Sigma^{-1} (\underline{\mu}_1 - \underline{\mu}_2).$$

Distance b/w two population $N(\underline{\mu}_1, \Sigma), N(\underline{\mu}_2, \Sigma)$

$$\underline{x} = (x_1, \dots, x_p)'$$

$$\underline{y} = (y_1, \dots, y_p)', \in \mathbb{R}^p.$$

$$d_s(\underline{x}, \underline{y}) = \sqrt{(\underline{x} - \underline{y})' \Sigma^{-1} (\underline{x} - \underline{y})}.$$

Properties.

(i) $d_s(\underline{x}, \underline{y}) = d_s(\underline{y}, \underline{x}).$

(ii) $d_s(\underline{x}, \underline{y}) > 0$ if $\underline{x} \neq \underline{y}$

(iii) $d_s(\underline{x}, \underline{y}) = 0$ if $\underline{x} = \underline{y}$

(iv) $d_s(\underline{x}, \underline{y}) \leq d_s(\underline{x}, \underline{z}) + d_s(\underline{z}, \underline{y}).$

↳ [Triangle Inequality]

RAO'S U STATISTIC

$$\underline{Y} \sim N(\underline{\mu}, \Sigma), A \sim W(n, \Sigma)$$

independent.

Partition

$$\underline{Y} = \begin{pmatrix} \underline{y}_1^{r \times 1} \\ \underline{y}_2^{p-r \times 1} \end{pmatrix}$$

$$\underline{\mu} = \begin{pmatrix} \underline{\mu}_1 \\ \underline{\mu}_2 \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

Consider $\Delta_p^2 = \underline{Y}' \Sigma^{-1} \underline{Y}$

$$= (\underline{\mu}_1' \quad \underline{\mu}_2') \begin{pmatrix} \Sigma_{11}^{-1} + \beta' \Sigma_{22}^{-1} \beta & -\beta \Sigma_{22}^{-1} \\ -\Sigma_{22}^{-1} \beta & \Sigma_{22}^{-1} \end{pmatrix} \begin{pmatrix} \underline{\mu}_1 \\ \underline{\mu}_2 \end{pmatrix}$$

$$= \underline{\mu}_1' \Sigma_{11}^{-1} \underline{\mu}_1 + (\underline{\mu}_2 - \beta \underline{\mu}_1)' \Sigma_{22}^{-1} (\underline{\mu}_2 - \beta \underline{\mu}_1)$$

$$\Delta_p^2 = \Delta_q^2 + \dots$$

$$(C, C) \cdot ab = (C, C) \cdot ab \quad (i)$$

$$C + x \cdot i \cdot C = (C, C) \cdot ab \quad (ii)$$

$$C = x \cdot i \cdot C = (C, C) \cdot ab \quad (iii)$$

$$(C, C) \cdot ab + (C, C) \cdot ab = (C, C) \cdot ab \quad (iv)$$

separating

RAO'S U STATISTIC

$$\underline{Y} \sim N(\underline{\mu}, \Sigma), A \sim W(n, \Sigma).$$

independent.

Partition

$$\underline{Y} = \begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

$$\underline{\mu} = \begin{pmatrix} \underline{\mu}_1 \\ \underline{\mu}_2 \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

Consider $\Delta_p^2 = \underline{Y}' \Sigma^{-1} \underline{Y}$

$$= \begin{pmatrix} \underline{\mu}_1' & \underline{\mu}_2' \end{pmatrix} \begin{pmatrix} \Sigma_{11}^{-1} + \beta' \Sigma_{22}^{-1} \beta & -\beta \Sigma_{22}^{-1} \\ -\Sigma_{22}^{-1} \beta & \Sigma_{22}^{-1} \end{pmatrix} \begin{pmatrix} \underline{\mu}_1 \\ \underline{\mu}_2 \end{pmatrix}$$

$\beta = \Sigma_{21} \Sigma_{11}^{-1}$ (?)

$$= \underline{\mu}_1' \Sigma_{11}^{-1} \underline{\mu}_1 + (\underline{\mu}_2 - \beta \underline{\mu}_1)' \Sigma_{22.1}^{-1} (\underline{\mu}_2 - \beta \underline{\mu}_1)$$

$$\Delta_p^2 = \Delta_q^2 + (\underline{\mu}_2 - \beta \underline{\mu}_1)' \Sigma_{22.1}^{-1} (\underline{\mu}_2 - \beta \underline{\mu}_1)$$

~~05/02/2024~~ 05/02/2024

$$T_p^2 = \underline{Y}' A^{-1} \underline{Y} = \begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \end{pmatrix}' \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \end{pmatrix}$$

$$= T_q^2 + (\underline{y}_2 - B \underline{y}_1)' A_{22.1}^{-1} (\underline{y}_2 - B \underline{y}_1)$$

$$[B = A_{21} A_{11}^{-1}]$$

$$\Rightarrow \frac{T_p^2}{n} = \frac{T_q^2}{n} + \underline{\underline{z'}} \underline{\underline{A_{22}^{-1}}} \underline{\underline{z}} \quad [n = N-1]$$

Define,
$$U = \frac{1 + T_q^2/n}{1 + T_p^2/n} \rightarrow \text{Rao's } U\text{-statistic.}$$

and,
$$\frac{n-p+1}{p-q} \cdot \left(\frac{1}{U} - 1 \right) \sim F(p-q, n-p+1) \text{ when } \Delta_p^2 = \Delta_q^2$$

$$\left\{ \begin{array}{l} \Delta_p^2 = \text{Distance b/w 2 populations with all } p \text{ variables} \\ \Delta_q^2 = \text{Distance b/w 2 populations with all } q \text{ variables.} \end{array} \right. \quad [p > q]$$

The null-hypothesis: $H_0: \Delta_p^2 = \Delta_q^2$

First q factors are equally informative.

Last $(p-q)$ factors are not useful (redundant).

Now, $\Delta_p^2 = \Delta_q^2$ if $\mu_2 - \beta \mu_1 = 0$ or $\underline{\underline{\Sigma_{22}^{-1}}} = 0$.
this is not possible

Hence, $\Delta_p^2 = \Delta_q^2$ if $\underline{\underline{\mu_2}} = \underline{\underline{\beta \mu_1}}$

$\rightarrow$ If $\underline{\underline{\mu_1}} = \underline{\underline{0}}$, then $\underline{\underline{\mu_2}} = \underline{\underline{0}}$.

So, the null hypothesis can be reformulated as -

$H_0: \underline{\underline{\mu_1}} = \underline{\underline{0}} \text{ (if } \underline{\underline{\mu_2}} = \underline{\underline{0}} \text{)}$

$\rightarrow$ The linear combination of $\underline{\underline{x}}$ $\underline{\underline{a'}} \underline{\underline{x}} \quad [\underline{\underline{x}} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}]$
Such that $\underline{\underline{x}}$ & $\underline{\underline{x_1}}$ are uncorrelated.

Hence, $\text{cor}(\underline{\underline{a'}} \underline{\underline{x}}, \underline{\underline{x_1}}) = 0$

Or, $\text{cor}(\underline{\underline{a_1'}} \underline{\underline{x_1}} + \underline{\underline{a_2'}} \underline{\underline{x_2}}, \underline{\underline{x_1}}) = 0$

$= \underline{\underline{a_1'}} \underline{\underline{\Sigma_{11}}} + \underline{\underline{a_2'}} \underline{\underline{\Sigma_{21}}} = 0 \Rightarrow \underline{\underline{a_1'}} = -\underline{\underline{a_2'}} \underline{\underline{\beta}}$

$$E(\underline{a}' \underline{x}) = \underline{a}' \underline{\mu}_1 + \underline{a}' \underline{\mu}_2 \\ = \underline{a}' (\underline{\mu}_2 - \beta \underline{\mu}_1)$$

$$\text{If } E(\underline{a}' \underline{x}) = 0 \quad \forall \underline{a}_1$$

$$\Rightarrow \underline{\mu}_2 - \beta \underline{\mu}_1 = 0$$

$$\Rightarrow \underline{\mu}_2 = \beta \underline{\mu}_1$$

06/02/2024

MULTIVARIATE BEHRENS - FISHER PROBLEM

$$H_0: \underline{\mu}_1 = \underline{\mu}_2$$

We have two independent normal population - one from $N(\underline{\mu}_1, \Sigma_1)$, $N(\underline{\mu}_2, \Sigma_2)$

Σ_1, Σ_2 : unknown & unequal.

Sample: $\underline{x}_\alpha^{(1)}, \alpha = 1(1)N_1$

$\underline{x}_\alpha^{(2)}, \alpha = 1(1)N_2$

So,

$$\underline{\bar{x}}_1 - \underline{\bar{x}}_2 \sim N\left(\underline{\mu}_1 - \underline{\mu}_2, \frac{\Sigma_1}{N_1} + \frac{\Sigma_2}{N_2}\right)$$

Sample
Covariance
matrix

$$\sum_{\alpha=1}^{N_1} (\underline{x}_\alpha^{(1)} - \underline{\bar{x}}_1)(\underline{x}_\alpha^{(1)} - \underline{\bar{x}}_1)' \\ + \sum_{\alpha=1}^{N_2} (\underline{x}_\alpha^{(2)} - \underline{\bar{x}}_2)(\underline{x}_\alpha^{(2)} - \underline{\bar{x}}_2)'$$

But this doesn't have Wishart distribution with covariance matrix multiplied of $\frac{\Sigma_1}{N_1} + \frac{\Sigma_2}{N_2}$

$$\text{Now, } \underline{y} \sim N\left(\underline{\mu} - \underline{\mu}_2, \Sigma^*\right)$$

If $\Sigma_1 = \Sigma_2$ then the test statistic will follow Hotelling's T^2

Case-I $N_1 = N_2 = N$

$$H_0: \underline{\mu}_1 = \underline{\mu}_2 \text{ or } \underline{\mu}_1 - \underline{\mu}_2 = \underline{\mu}^* = 0$$

$$\text{Then } \underline{\bar{X}}_1 - \underline{\bar{X}}_2 \sim N(\underline{\mu}_1 - \underline{\mu}_2, \frac{\underline{\Sigma}_1}{N} + \frac{\underline{\Sigma}_2}{N})$$

$$\text{Define, } \underline{Y}_\alpha = \underline{X}_\alpha^{(1)} - \underline{X}_\alpha^{(2)} \rightarrow \underline{\mu}^* \rightarrow \underline{\Sigma}^*$$

$$\sim N(\underline{\mu}_1 - \underline{\mu}_2, \underline{\Sigma}_1 + \underline{\Sigma}_2)$$

independent of $\underline{Y}_\beta (\beta \neq \alpha)$

$$\text{Let, } \underline{\bar{Y}} = \frac{1}{N} \sum_{\alpha=1}^N \underline{Y}_\alpha, \quad \underline{S} = \frac{1}{N-1} \sum_{\alpha=1}^N (\underline{Y}_\alpha - \underline{\bar{Y}})(\underline{Y}_\alpha - \underline{\bar{Y}})'$$

$$\text{Therefore, } \boxed{T^2 = N \underline{\bar{Y}}' \underline{S}^{-1} \underline{\bar{Y}}}$$

$$\underline{T}^2 \sim \frac{T^2}{N-1}$$

But in ~~the~~ Fisher-T setup of univariate we had $n_1 + n_2 - 2$ d.f. Similarly, we ~~to~~ should have $N + N - 2 = 2(N-1)$ d.f. But we are having $N-1$ d.f. i.e. there is a loss of $N-1$ d.f.

[t-distn with higher df $\Rightarrow$ thinner tail]

$\Rightarrow$ less type-I error

So, here, probability of rejecting true H_0 becomes higher.]

Case-II $N_1 \neq N_2$

$$\underline{Y}_\alpha = \left(\underline{X}_\alpha^{(1)} - \sqrt{\frac{N_1}{N_2}} \underline{X}_\alpha^{(2)} \right) + \frac{1}{\sqrt{N_1 N_2}} \left(\sum_{\beta=1}^{N_1} \underline{X}_\beta^{(1)} - \sqrt{\frac{N_1}{N_2}} \sum_{\gamma=1}^{N_2} \underline{X}_\gamma^{(2)} \right)$$

$$= \left(\underline{X}_\alpha^{(1)} - \sqrt{\frac{N_1}{N_2}} \underline{X}_\alpha^{(2)} \right) + \left(\frac{1}{\sqrt{N_1 N_2}} \sum_{\beta=1}^{N_1} \underline{X}_\beta^{(1)} - \frac{1}{\sqrt{N_2}} \sum_{\gamma=1}^{N_2} \underline{X}_\gamma^{(2)} \right)$$

$$E(\underline{Y}_\alpha) = \underline{\mu}_1 - \sqrt{\frac{N_1}{N_2}} \underline{\mu}_2 + \frac{N_1}{\sqrt{N_1 N_2}} \underline{\mu}_1 - \frac{N_2}{\sqrt{N_1 N_2}} \underline{\mu}_2$$

$$= \underline{\mu}_1 - \underline{\mu}_2 = \underline{\mu}^*$$

$$\begin{aligned} \text{Cov}(\underline{Y}_\alpha) &= E[(\underline{Y}_\alpha - E(\underline{Y}_\alpha))(\underline{Y}_\alpha - E(\underline{Y}_\alpha))'] \\ &= \left(\Sigma_1 + \frac{N_1}{N_2} \Sigma_2 \right) \delta_{\alpha\beta} \quad \text{where } \delta_{\alpha\beta} = \begin{cases} 1, & \text{if } \alpha = \beta \\ 0, & \text{otherwise} \end{cases} \\ &= \Sigma^* \end{aligned}$$

So, $H_0: E(\underline{Y}_\alpha) = \underline{0}$

The test statistic is -

$$T^2 = N_1 \bar{\underline{Y}}' S^{-1} \bar{\underline{Y}} \sim T_{N-1}^2 \text{ (under } H_0 \text{)}$$

where $\bar{\underline{Y}} = \frac{1}{N_1} \sum_{\alpha=1}^{N_1} \underline{Y}_\alpha$

$$\bar{\underline{Y}} = \bar{\underline{X}}_1 - \bar{\underline{X}}_2$$

$$(N_1 - 1) S = \sum_{\alpha=1}^{N_1} (\underline{Y}_\alpha - \bar{\underline{Y}})(\underline{Y}_\alpha - \bar{\underline{Y}})'$$

Let's take a more generalized variance,

$$H_0: \sum_{i=1}^q \beta_i \underline{\mu}_i = \underline{0}$$

where $\beta_1, \dots, \beta_q$ are given scalars.

$$\underline{X}_\alpha^{(i)} \sim N(\underline{\mu}_i, \Sigma_i), \quad i = 1, 2, \dots, q$$

$$\alpha = (i-1)N_i + j$$

$$\underline{Y}_\alpha = \beta_1 \underline{X}_\alpha^{(1)} + \sum_{i=2}^q \beta_i \left(\underline{X}_\alpha^{(i)} - \frac{1}{N_i} \sum_{j=1}^{N_i} \underline{X}_j^{(i)} \right) + \frac{1}{N_1 N_2} \sum_{j=1}^{N_1} \underline{X}_j^{(1)}$$

$$E(\underline{Y}_\alpha) = \sum \beta_i \underline{\mu}_i$$

$$V(\underline{Y}_\alpha) = \left(\sum_{i=1}^q \frac{\beta_i^2 N_i}{N_1} \Sigma_i \right) \delta_{\alpha\beta}$$

Under H_0 , test statistic will follow -